SPE 9723

# Comparison of Solutions to a Three-Dimensional Black-Oil Reservoir Simulation Problem

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## Summary

A comparison of solutions to a three-dimensional black-oil reservoir simulation problem is presented. The test of the problem and a brief description of the seven simulators used in the study are given.


## Introduction

Seven companies participated in a reservoir simulation project to compare the results obtained by different black-oil simulators. The companies were chosen to give a good cross section of the solution methods used in the industry. The participants were Amoco Production Co., Computer Modelling Group of Calgary (CMG), Exxon Production Research Co., Intercomp Resource Development and Engineering Inc., Mobil Research and Development Corp., Shell Development Co., and Scientific Software Corp. (SSC). The paper presents the text of the problem, a comparison of results in graphical form, and a brief description of each model. The descriptions were supplied by the participants.

A variety of computers was used. Amoco used IBM 3033, IBM 370/168, and Amdahl V/6. CMG used Honeywell 6000 DPS, and Exxon used Amdahl 470/V5 and IBM 370/168. Intercomp used Cray-1 and Harris/7. Mobil and SSC used CDC Cyber 175, and Shell used Univac 1110/2C Level 36. The number of time steps and the central processor times varied considerably. Those interested in the actual values should contact the individual companies.

Except for Shell, all the participants used single-point upstream mobility weighting. Shell used two points upstream. Constraints and data are given in the text.

## Statement of the Problem

Areal and cross-section views of the reservoir are given in Figs. 1 and 2. The grid system is given in Fig. 1. Stratification and reservoir properties are given in Fig. 2. The reservoir is initially undersaturated. A gas injection well is located at Grid Point (1, 1), and a producing well is located at Grid Point (10, 10). Pertinent data and constraints are given in Table 1. PVT properties and relative permeabilities are given in Tables 2 and 3. The participants were asked to make the runs and report the results described below.

## Runs To Be Made

### Case 1

Let the bubble-point (saturation) pressure be constant with a value equal to the original value.

### Case 2

Let the saturation pressure vary with gas saturation – i.e., this is a variable saturation-pressure case. The PVT lines at pressures above the calculated saturation pressures are parallel to the original line.

## Results To Be Reported

The following results are to be reported.

1. Plots of:
   a. Oil rate vs. time.
   b. GOR vs. time.
2. Report annually and at abandonment:
   a. The pressures of the cell where the injector and producer are located.*
   b. Gas saturation at Grid Points (1, 1, 1), (1, 1, 2), (1, 1, 3), (10, 1, 1), (10, 1, 2), (10, 1, 3), (10, 10, 1), (10, 10, 2), and (10, 10, 3).
3. Report at the end of 8 years:
   a. Tables of gas saturation.
   b. Tables of cell pressures.*
   c. Tables of saturation pressures for the variable saturations-pressure case.*

*Report of all pressures referred to a depth of 8,325 ft or at the center of the respective blocks. If both are available, report both.

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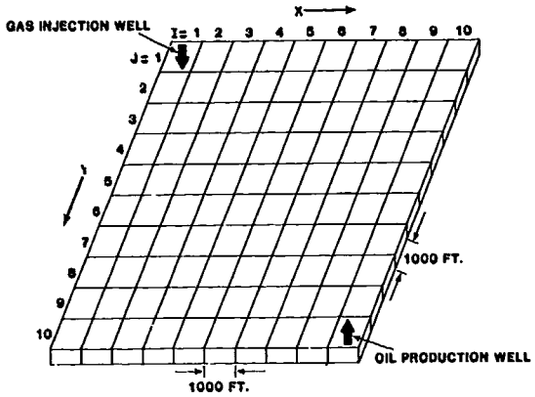


Fig. 1 – Reservoir and grid system.

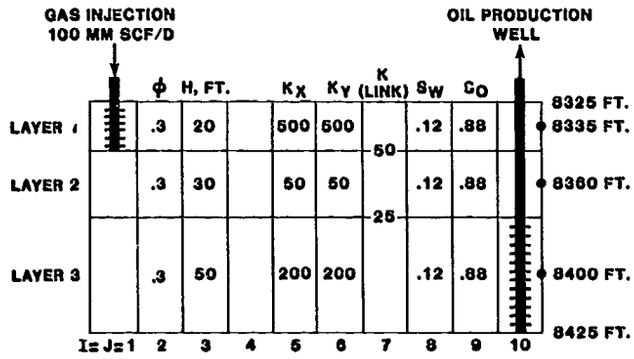


Fig. 2 – Diagonal cross section.

## TABLE 1 – DATA AND CONSTRAINTS

| | |
|---|---|
| Initial reservoir pressure, psia at 8,400 ft | 4,800 |
| Gas injection rate, MMscf/D | 100 |
| Maximum oil production rate, STB/D | 20,000 |
| Minimum oil rate, STB/D | 1,000 |
| Minimum flowing bottomhole pressure, psi | 1,000 |
| Maximum saturation change during time step | 0.05 |
| Rock compressibility, 1/psi | $3 \times 10^{-6}$ |
| Porosity value of 0.3 was measured at a base pressure of 14.7 psi | |
| Wellbore radius, ft | 0.25 |
| Skin | 0 |
| Capillary pressure | 0 |
| Reservoir temperature, &F | 200 |
| Gas specific gravity | 0.792 |

Runs are terminated either at the end of 10 years or when GOR≤20,000 scf/STB or when the oil production rate≤1,000 STB/D; whichever occurs first terminates the run.

## TABLE 2 – PVT PROPERTIES

Saturated Oil PVT Functions

| Reservoir Pressure (psia) | FVF (RB/STB) | Viscosity (cp) | Density (lbm/cu ft) | Solution GOR (scf/stb) |
|---|---|---|---|---|
| 14.7 | 1.0620 | 1.0400 | 46.244 | 1.0 |
| 264.7 | 1.1500 | 0.9750 | 43.544 | 90.5 |
| 514.7 | 1.2070 | 0.9100 | 42.287 | 180.0 |
| 1014.7 | 1.2950 | 0.8300 | 41.004 | 371.0 |
| 2014.7 | 1.4350 | 0.6950 | 38.995 | 636.0 |
| 2514.7 | 1.5000 | 0.6410 | 38.304 | 775.0 |
| 3014.7 | 1.5650 | 0.5940 | 37.781 | 930.0 |
| 4014.7 | 1.6950 | 0.5100 | 37.046 | 1270.0 |
| 5014.7 | 1.8270 | 0.4490 | 36.424 | 1618.0 |
| 9014.7 | 2.3570 | 0.2030 | 34.482 | 2984.0 |

Saturated Water PVT Functions

| Reservoir Pressure (psia) | FVF (RB/bbl) | Viscosity (cp) | Density (lbm/cu ft) | Gas/Water Ratio (scf/bbl) |
|---|---|---|---|---|
| 14.7 | 1.0410 | 0.3100 | 62.238 | 0.0 |
| 264.7 | 1.0403 | 0.3100 | 62.283 | 0.0 |
| 514.7 | 1.0395 | 0.3100 | 62.328 | 0.0 |
| 1014.7 | 1.0380 | 0.3100 | 62.418 | 0.0 |
| 2014.7 | 1.0350 | 0.3100 | 62.599 | 0.0 |
| 2514.7 | 1.0335 | 0.3100 | 62.690 | 0.0 |
| 3014.7 | 1.0320 | 0.3100 | 62.781 | 0.0 |
| 4014.7 | 1.0290 | 0.3100 | 62.964 | 0.0 |
| 5014.7 | 1.0258 | 0.3100 | 63.160 | 0.0 |
| 9014.7 | 1.0130 | 0.3100 | 63.959 | 0.0 |

Undersaturated Oil PVT Functions

| Reservoir Pressure (psia) | FVF (RB/STB) | Viscosity (cp) | Density (lbm/cu ft) |
|---|---|---|---|
| 4014.7 | 1.6950 | 0.5100 | 37.046 |
| 9014.7 | 1.5790 | 0.7400 | 39.768 |

Undersaturated Water PVT Functions

| Reservoir Pressure (psia) | FVF (RB/bbl) | Viscosity (cp) | Density (lbm/cu ft) |
|---|---|---|---|
| 4014.7 | 1.0290 | 0.3100 | 62.964 |
| 9014.7 | 1.0130 | 0.3100 | 63.959 |

Gas PVT Functions

| Reservoir Pressure (psia) | FVF (RB/bbl) | Viscosity (cp) | Density (lbm/cu ft) | Pseudo Gas Potential $M(p)$ (psia$^2$/cp) |
|---|---|---|---|---|
| 14.7 | 0.166666 | 0.008000 | 0.0647 | 0. |
| 264.7 | 0.012093 | 0.009600 | 0.8916 | 0.777916 E+07 |
| 514.7 | 0.006274 | 0.011200 | 1.7185 | 0.267580 E+08 |
| 1014.7 | 0.003197 | 0.014000 | 3.3727 | 0.875262 E+08 |
| 2014.7 | 0.001614 | 0.018900 | 6.6806 | 0.270709 E+09 |
| 2514.7 | 0.001294 | 0.020800 | 8.3326 | 0.386910 E+09 |
| 3014.7 | 0.001080 | 0.022800 | 9.9837 | 0.516118 E+09 |
| 4014.7 | 0.000811 | 0.026800 | 13.2952 | 0.803963 E+09 |
| 5014.7 | 0.000649 | 0.030900 | 16.6139 | 0.112256 E+10 |
| 9014.7 | 0.000386 | 0.047000 | 27.9483 | 0.251845 E+10 |

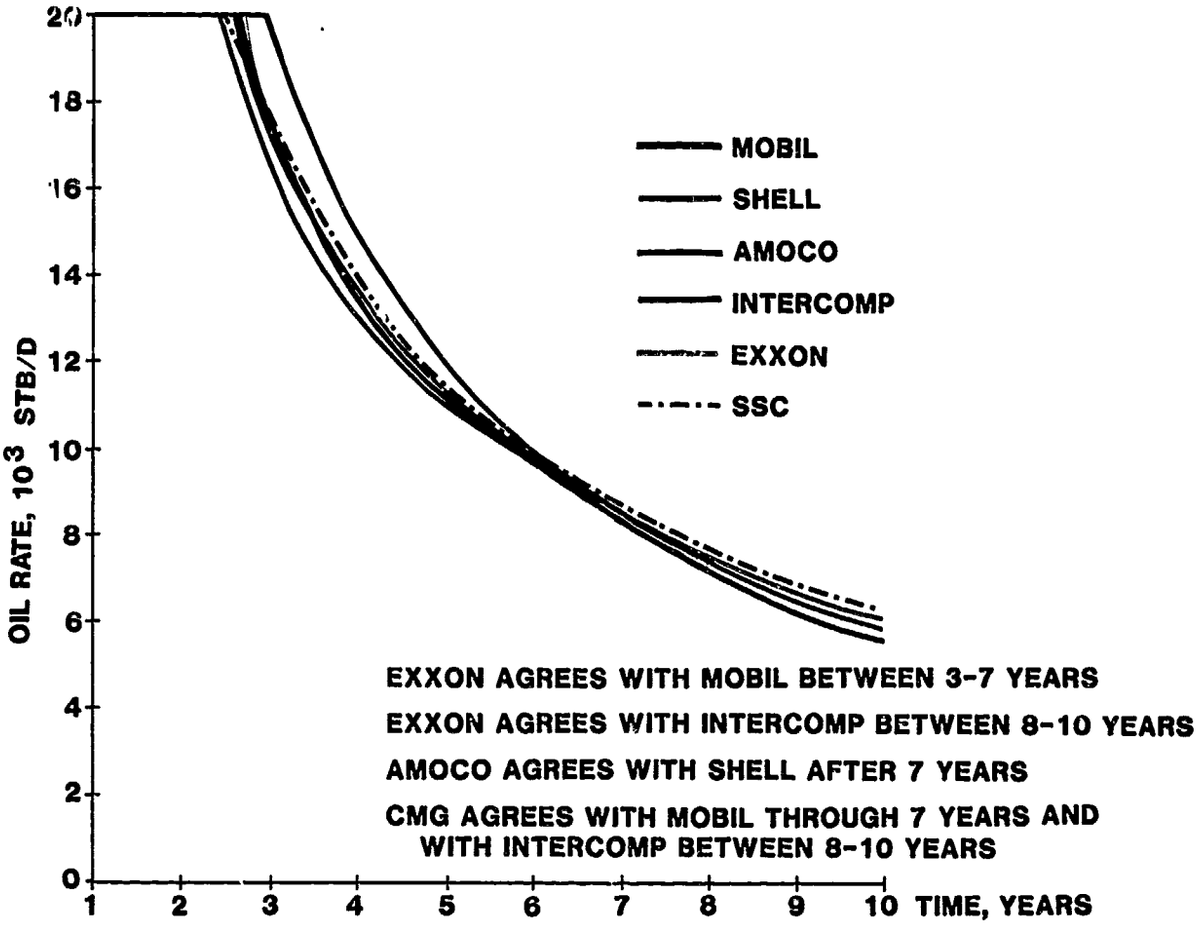


Fig. 3 – Case 1 – oil rate vs. time.

## Results

A comparison of the results is given in Figs. 3 through 18. No comparison of saturation pressures is given because the values reported by the seven companies were within 20 psi of each other.

## Description of the Simulators

### Amoco's Model

The IMPES method was used, with semi-implicit adjustments in well rates. This method proved quite satisfactory; additional computations for implicit handling of interblock flow were not needed.

Maximum time-step size can vary with time and is input. The model determines internal time-step sizes to satisfy both the current maximum $\Delta t$ and the maximum saturation change for any grid block (5% PV). A sequence of runs using maximum $\Delta t$ of 0.25, 0.5, 1, and 2 months yielded virtually identical results, confirming the applicability of the IMPES method. The final results are for a maximum $\Delta t$ of 1.0 month.

For each internal time step the computation sequence was as follows.

1. Well rates.
2. Coefficients including terms for semi-implicit production rates.
3. Iterative computation of grid-block pressure

TABLE 3 – RELATIVE PERMEABILITY DATA*

| | Oil-Gas | |
|---|---|---|
| $s_g$ | $k_{rg}$ | $k_{ro}$ |
| 0 | 0.0 | 1.0 |
| 0.001 | 0.0 | 1.0 |
| 0.02 | 0.0 | 0.997 |
| 0.05 | 0.005 | 0.980 |
| 0.12 | 0.025 | 0.700 |
| 0.2 | 0.075 | 0.350 |
| 0.25 | 0.125 | 0.200 |
| 0.3 | 0.190 | 0.090 |
| 0.4 | 0.410 | 0.021 |
| 0.45 | 0.60 | 0.010 |
| 0.5 | 0.72 | 0.001 |
| 0.6 | 0.87 | 0.0001 |
| 0.7 | 0.94 | 0.000 |
| 0.85 | 0.98 | 0.000 |
| 1.0 | 1.0 | 0.000 |

*This is a two-phase, gas/oil problem. Set the relative permeability to water equal to zero for all values of water saturations.